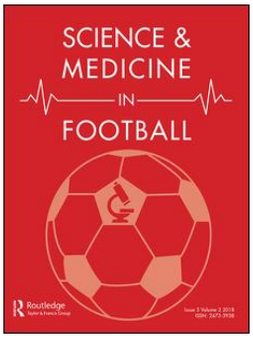

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ARTICLE

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Workload monotony, strain and non-contact injury incidence in professional football players

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ABSTRACT

Objectives: The purpose of this study was to analyse the association of the workload monotony and workload strain with the injury incidence with professional football players taking part in European football competitions.
Methods: A total of 130 elite football players, from 5 teams playing European competitions, have been followed during a season. The internal workload was calculated using the Session Rate of Perceived Exertion method and the injuries were recorded. The 4-weeks, 3-weeks, 2-weeks and 1-week monotony (mean workload divided by workload standard deviation) and strain (absolute workload multiplied by monotony) were calculated on a daily basis, using a rolling days method.
Results: The 4-weeks monotony was related to injury incidence. The relative risk (RR) of injury is decreased with an increase in 4-weeks monotony (RR = 0.72, 95% confidence interval [CI] 0.58–0.90; $P = 0.004$). The injury incidence was related to the 4-weeks strain. The RR of injury is increased with an increase 4-weeks strain (RR = 1.50, 95% CI 1.14–1.97; $P = 0.004$).
Conclusions: The links between a high 4-weeks monotony and the injury incidence, and the links between a high 4-weeks strain and the injury incidence indicate that monotony and strain were associated with injury incidence in elite football players and require to be monitored.

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KEYWORDS
Team sport; training; injury prevention

Introduction

Although football has significant beneficial effects on health profile and physical capacity, it also presents some risks, such as injury (Krustrup et al. 2010). Injuries in elite professional football are a major concern for teams as the overall risk is extremely high: about 1000 times higher than for high-risk industrial occupations (Drawer and Fuller 2002). On average, a player sustains two injuries per season, resulting in approximately 50 injuries per team and season (Ekstrand et al. 2011).

The workload has been identified as an injury risk factor among several activities (Jones et al. 2017). This result highlights the importance of monitoring players in a practical setting. Several studies associated workload with injury incidence among professional football players (Malone et al. 2017; Lu et al. 2017; Jaspers et al. 2017), confirming the importance of monitoring players workload in order to prevent injury.

Two reviews of literature concluded that internal/subjective tools were more sensitive and related to injury incidence than external and/or objective tools (Saw et al. 2016; Jones et al. 2017). The main subjective tool used by practitioners to calculate the internal workload is the session Rate of Perceived Exertion (sRPE), using a modified perception scale (Borg, CR-10) (Akenhead and Nassis 2016). Workload is calculated using the method proposed by Foster (1998), whereby the session intensity rating is multiplied by the training/match duration (min). A link between the absolute workload calculated with the sRPE and injury incidence has been established in football. Malone et al. (2017) found that a weekly workload higher than 1500 arbitrary units (A.U.) was associated with an increased injury incidence in elite football players. In the original study, Foster (1998) reported a link between spikes in the workload monotony, strain and illness. The monotony is defined as the daily mean workload divided by the daily workload standard deviation (Foster 1998). In this study of 1998, Foster showed that 77% of illnesses could be explained by a spike in training load monotony. The strain is defined as the monotony multiplied by the workload (Foster 1998). In this study, 89% of illnesses could be explained by a spike in training load strain. Based on these results, it seems that an elevated monotony or an elevated strain in workload could have negative impacts on athletes' health and training adaptations. One study identified a link between monotony, strain and injury incidence among young football players (Brink et al. 2010). In this study, increase in monotony (odds ratio [OR] = 2.59, 95% confidence interval [CI] 1.22–1.50) or strain (OR = 1.01, 95% CI 1.00–1.02) was significantly related to an increase in injury incidence. However, this study has been held with a youth academy and, to our knowledge, no study evaluated the link between these original indicators and the injury incidence with professional football players. Football players playing European competitions are requested to play 2–3 games in a week, leading to variations in the workload. This congested match schedule modifies monotony and strain. The monotony and strain could

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be interesting indicators to monitor workload with professional football players playing European competitions to reduce injury incidence.

Therefore, the aim of this study is to evaluate the link between workload monotony, strain and injury incidence with elite football players taking part in European competitions.

Materials and methods

Participants

A total of 130 professional football players (age: 26.3 ± 4.9 years; height: 181.9 ± 7.4 cm; body mass: 77.6 ± 7.4 kg) from 5 different European professional football clubs playing in top leagues (2 in French Ligue 1, 1 in Spanish Primera Liga, 1 in Italian Serie A and 1 in the English Premier League) playing at the highest level of European competitions (UEFA Champion's League and Europa League) were followed during a full-season (2014–2015). All players were informed and consented to take part in the study.

All players from the first-team squad were included in the study. If a player joined the team during the season, he was included from the date he joined the team. A player who left the team during the season was excluded from the study from the date he left the team. At the beginning of the observational period, 109 players were observed. Totally, 21 players were included later after joining one of the clubs, while 18 players were lost to follow after joining or leaving one of the clubs during the transfer windows. If a player was already injured at the start of data collection, he was included in the study, but this injury was excluded (Fuller et al. 2006). This study was conducted in accordance with the local ethical committee on biomedical research (CCTIRS#10,544) and the standards of the declaration of Helsinki.

Injury data collection

An injury was defined as any physical complaint sustained by a player that resulted from a football match or football training, that made the player unable to participate in future football training or a match (Fuller et al. 2006). In this study, only non-contact injuries were recorded, as these injuries represent the majority (of injury) in elite football (between 57% and 80%) (Ekstrand et al. 2011) and are largely preventable, contrary to contact injuries, which are considered as harder to prevent (Gabbett 2010). The non-contact injuries were diagnosed and recorded by the medical doctor working in the club. The injury incidence was calculated as the number of non-contact injuries per 1000 h of exposure. The period of exposition was recorded and the number of injuries were normalized for 1000 h of exposition.

Quantification of workload

The workload was calculated by the sRPE method. Players were instructed to rate the global intensity of all sessions and matches using a modified category ratio scale going from 0 to 10, based on the scale developed by Borg et al. (1987) by answering the following question: "How was your workout?". The scale was graduated unit by unit and with the following correspondence: 0 = rest, 1 = very weak, 2 = weak, 3 = moderate, 4 = somewhat hard, 5 = hard, 7 = very hard, 10 = maximal. The sRPE was collected 30 min after completion of the match/session (gym, bike, football session, injury prevention, etc.) by the same science or medicine department staff member. The players were isolated to answer in order not to be influenced by other players. Workload, expressed in arbitrary units (A.U.) was calculated by multiplying the perceived intensity by the session or match duration (Foster 1998). All training sessions and all matches were included. The 1-week monotony corresponded to the mean workload of the last 7 days divided by the standard deviation of workload of the last 7 days. To calculate the workload strain, the weekly monotony (7 days) was multiplied by the absolute workload of the last 7 days. The 2-weeks (14 last days), 3-weeks (21 last days) and 4-weeks (28 last days) monotony and strain were also assessed. The monotony and strain were calculated on a daily basis, using a rolling days method and the association with injuries was analysed on a daily basis.

Statistical analysis

A Poisson regression analysis was performed with IBM SPSS Statistics Version 20 (IBM Corp, Armonk, NY, USA) for the weekly, 2-weeks, 3-weeks and 4-weeks monotony and strain to assess the link between these factors and injury incidence and calculate relative risk (RR). The monotony and strain scores were log-transformed when the scores were not normally distributed. The level of significance was set at $P \leq 0.05$.

Descriptive statistics concerning the exposition, injury incidences, injury probability and workload were calculated with Excel (Microsoft Corporation, Redmond, WA, USA) (Version 16.6.7 for Mac). The study power to detect a 1.2 RR was 0.50, and with the sample size obtained here, this study has a 0.80 power to detect a RR of 1.38. The minimal sample size to detect a RR of 1.5 was 22,222 days of observation.

Results

During the observational period, 237 non-contact injuries were sustained; injury incidence was 7.4 injuries per 1000 h of exposure. This incidence corresponds to a probability of sustaining a non-contact injury during a training day of 0.78%, or a non-contact injury every 135 h of training. Among the 237 injuries, 58 injuries were categorized as re-injuries.

There was a total of 30,887 days of observation in this study.

The mean 4-weeks absolute workload was 7187 ± 3442 A.U., for 3-weeks 5496 ± 2,674 A.U., for 2 weeks 2737 ± 1295 A.U. and for 1-week 1914 ± 1080 A.U.

The Poisson regression indicated that there was no link between the weekly monotony and the injury incidence ($P = 0.43$), the 2-weeks ($P = 0.37$) and 3-weeks monotony ($P = 0.06$). The 4-weeks monotony was related to the injury incidence ($P < 0.004$). The regression analysis indicated that there was a decrease in the injury incidence with an increase of the 4-weeks monotony (RR = 0.72, 95% CI 0.58–0.90).

There was no link between the injury incidence and the weekly strain, the 2-weeks and the 3-weeks strain. The 4-weeks strain was related to the injury incidence (P = 0.004). The injury incidence was significantly increased when the 4-weeks strain was increased (= 1.50, 95% CI 1.14–1.97).

Discussion

The new findings in this study showed that an increase in monotony was associated with a decrease in injury incidence and that an increase in strain was associated with an increase in injury incidence.

In the current study, the overall non-contact injury incidence was 7.4 per 1000 h of exposure. These results confirm the results of previous studies concerning the injury incidence in elite professional football players (Ekstrand et al. 2013).

The link between injury and monotony has first been described by Foster (1998), and the monotony has been linked to illness episode with elite athletes in the original study about the sRPE to calculate the workload. The results concerning the monotony indicated that there was a link between the 4-weeks monotony and injury incidence. The results of the current study indicated that an elevated 4-weeks monotony was associated with a reduced risk of injury. The results of Foster (1998) and Brink et al. (2010) indicated that the illness and injury risk were increased when the weekly monotony was elevated. The differences in results could be explained by the differences in population (youth academy vs. professional players). The schedule are very different between youth academy and professional players, with more congested periods, with 3 games per week in professional football, which have a great impact on the workload monotony, while the schedule is more regular in youth teams. The differences in the results could also be explained by the length of the period used to calculate the monotony. In the present study, the link between monotony and injury was found with a 4-weeks period while it was only calculated for a 1-week period in Brink et al.'s (2010) study.

The results of the present study are surprising, because an increase in monotony was associated with an increase in injury incidence in previous studies (Foster 1998; Brink et al. 2010), but in accord with results found in other studies about workload and field-based team sports about the acute:chronic ratios in which spikes and brutal changes in the workload (which corresponds to a decrease in monotony) are related to an increase in the injury incidence (Hulin et al. 2016; Bowen et al. 2017; Malone et al. 2017). The results of the current study confirm that the workload consistency is a factor related to injury incidence.

This study has been held with elite football players taking part in European competitions (UEFA Champions League and Europa League). These 3 games weeks lead to a high workload and a high day-to-day workload variations, as the players probably sustain 2 spikes in the daily workload surrounded by very light workload days (recovery sessions). These weeks lead to very high daily variations and decrease the monotony. The results of the present study indicate that when these daily variations are repeated in the time (4 weeks), this is associated with an increase in the injury risk.

The second aim of the study was to analyse the link between the workload strain, which is the product of the absolute workload and the monotony. The study of Foster (1998) indicated that a high workload strain could lead to illness episodes with elite athletes. As for the monotony, this indicator has never been studied with professional football players. The first finding of the study is that the 4-weeks strain is related to the injury incidence, contrary to the 1-week, 2-weeks and 3-weeks strain. These results confirm the monotony results about the absence of link between these workload indicators and the injury incidence during a short period but the presence of link during longer periods.

The 4-weeks strain results confirm the ones found by Foster (1998), as a high strain is linked to an increased injury incidence. The players might not be able to cope with a high strain during long periods, which could explain the increase in injury risk with a high 4-weeks strain.

This study presents some limits. First of all, the workload was calculated with the sRPE. This tool represents the athletes' internal (total) workload based on his own perception of both physical strain and psychological stress. The subjectivity of this tool could be seen as a limit, as it requires that the athlete is familiarized with the sRPE, understands the tool and is honest when giving his sRPE score. However, the players were already familiarized with the tool and used to it and were frequently remembered the importance to answer honestly. This tool has been validated for use in football (Impellizzeri et al. 2004). Moreover, the internal and subjective tools to monitor the workload have been shown to be more sensitive and related to the injury incidence than external and/or objective tools (Saw et al. 2016; Jones et al. 2017). An isolated factor in a multifactorial context such as the football injury would have a very low ability to predict an injury. Therefore, predictive ability of the monotony and strain, which could be defined as the accuracy in predicting the occurrence (or not) of the injury with these factors has not been assessed. The present study is a cohort study and identified an association between workload monotony, strain and injury incidence. This association does not necessarily mean that there is a causal connection between monotony, strain and injury incidence (Bahr 2016). Randomised controlled trials should be assessed to analyse the effect of a controlled monotony and strain on the injury incidence, in an isolated way, to confirm a causal connection (Bahr 2016). Another limit of the study is that the workloads were recorded by sports scientists in the clubs who were aware of the workload sustained by the athletes and probably took into account these workloads to modify or adapt the sessions to have an impact on the future workloads.

Conclusion

To conclude, the results of this study show that a regular workload is an injury protective factor while a high workload strain (high and very regular workload) is a risk factor when it is sustained during a long period (4 weeks). These new findings highlight two new workload indicators that should be monitored by practitioners to reduce the injury incidence. These new indicators could be useful to prevent injury and could be assessed by practitioners in combination with other

factors to reduce the injury risk in the multifactorial context of injury in football. The results concerning the monotony should bring the sport scientists and practitioners to manage the day-to-day workload variations (caused by a gap between rest day workload, training day workload and spikes in the loads caused by games) to try to decrease the injury incidence.

Practical implications

The results of the present study indicate that the 4 weeks workload monotony and the 4 weeks workload strain are associated with an increase in injury incidence. These results highlight that these two indicators should be monitored in combination with other factors by practitioners to reduce the injury incidence in professional football players. The monotony results should bring the sport scientists and practitioners to reduce the day-to-day workload variations to try to reduce the injury incidence.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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